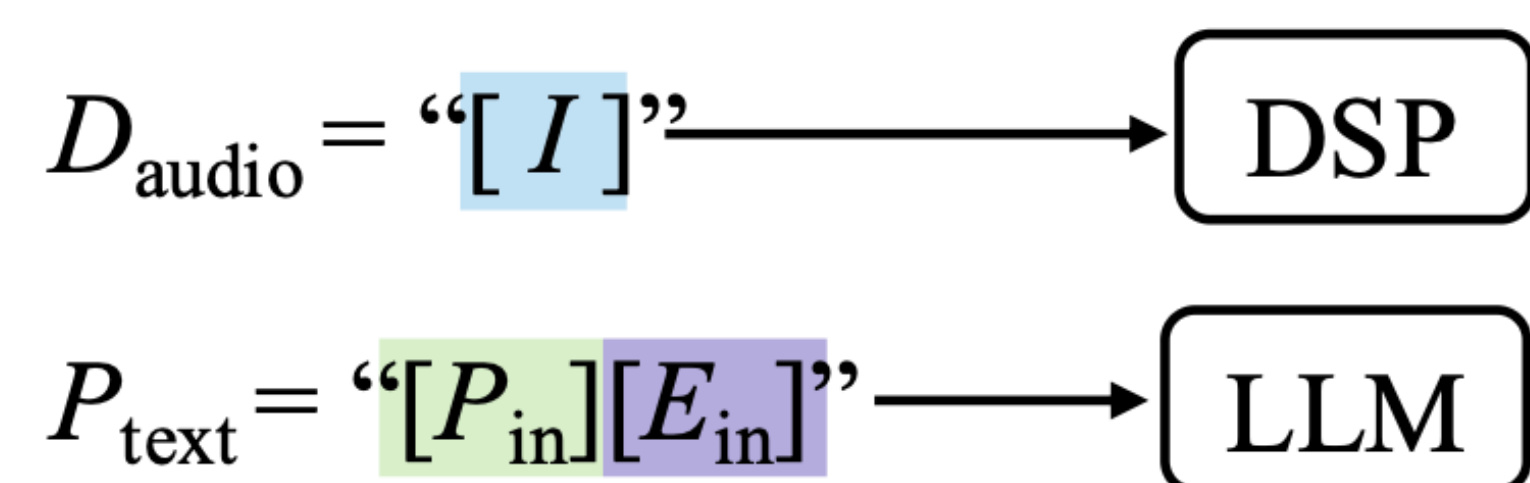
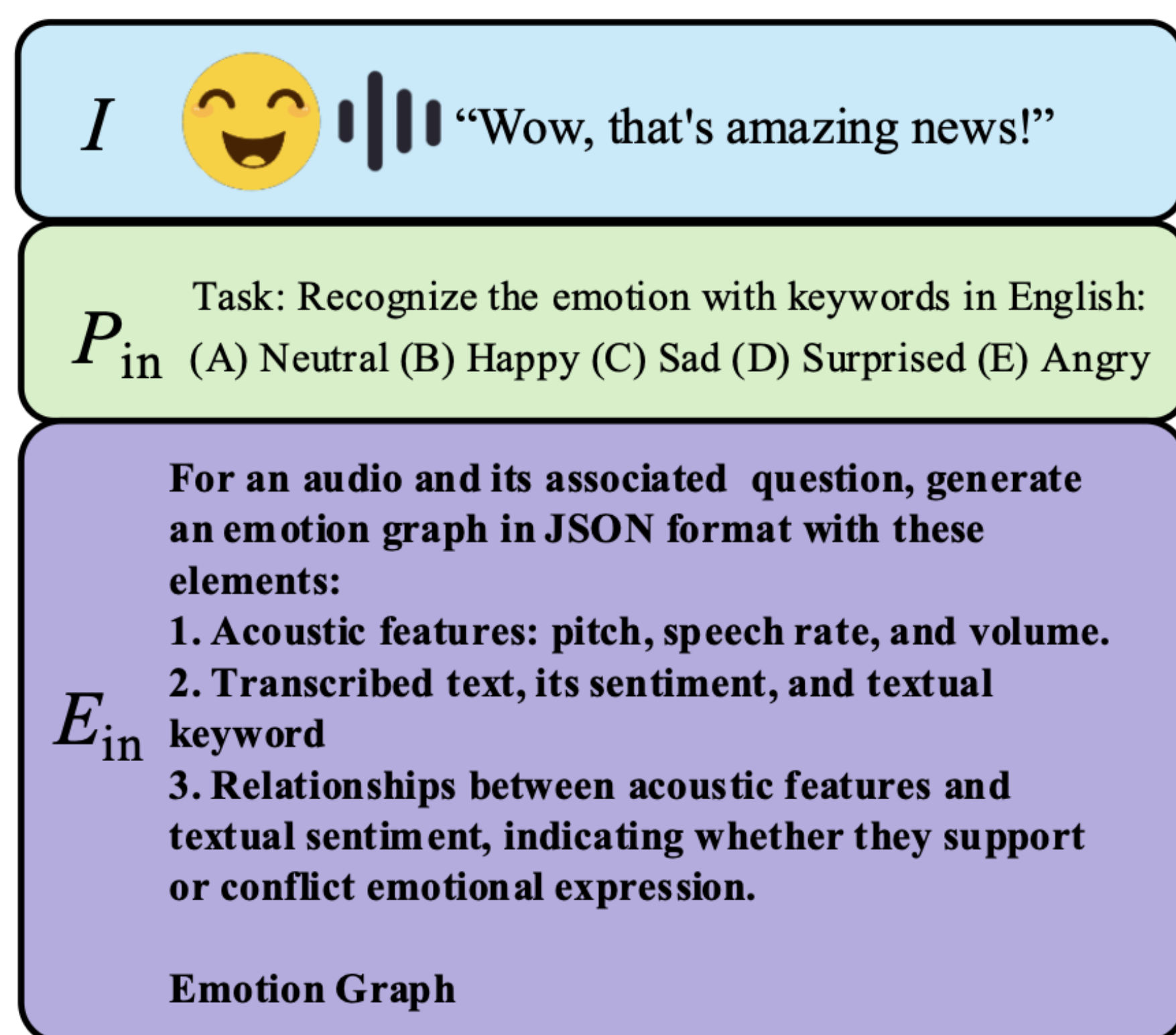


Motivation

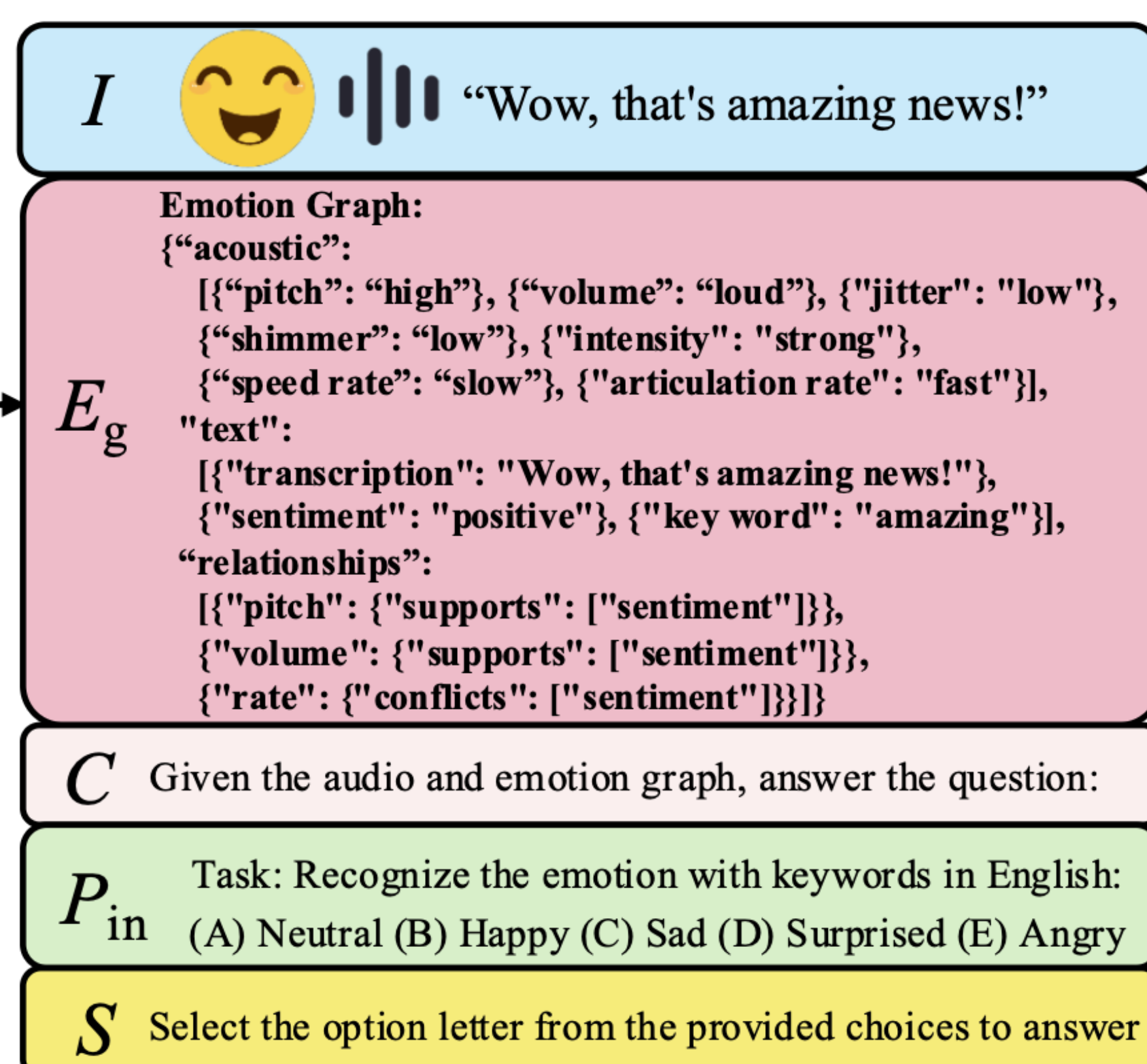
- LALMs focus on **linguistic semantics**, ignore **paralinguistic cues**
- Zero-shot SER lacks **structured reasoning**
- CoT prompting is **unstructured** → **hallucination-prone**

Method

1. Emotion Graph Generation



2. Response Generation



Key Idea: Replace free-form CoT with structured Emotion Graph reasoning

Conclusion

- **Structured EG** enables interpretable emotion reasoning
- **Plug-and-play** zero-shot improvement across LALMs
- **Structured > free-form CoT** for multimodal reasoning
- **Key Insight: Structure matters more than scale** for emotion reasoning

Experiments

Speech Emotion Recognition (Acc%)						
Method	IEMOCAP	MELD	ESD	MER test1	MER test2	Overall
<i>Encoder-based Classification Models</i>						
HuBERT-Large [17]	64.6	53.2	70.5	55.6	45.3	57.8
wav2vec2-Large [18]	69.3	54.8	64.0	41.2	40.6	54.0
WavLM-Large [19]	68.9	54.6	70.3	48.3	42.8	57.0
CLEP-DG [20]	74.7	55.2	69.4	54.8	49.5	60.7
<i>LLM-based Generative Models</i>						
Text+LLM	54.8	54.0	54.3	47.2	44.5	50.9
Whisper+LLM [8]	57.1	53.8	53.7	49.4	46.9	52.2
OSUM [10]	52.8	53.1	50.9	62.5	55.2	54.9
SALMONN-7B [9]	67.0	32.9	51.8	45.8	41.7	47.8
SenseVoice-L [21]	71.3	54.7	65.6	59.3	56.7	61.5
BLSP-Emo [7]	76.1	57.2	72.2	60.0	54.7	64.0
Qwen2-Audio [1]	65.5	55.5	57.1	52.9	47.4	55.8
Qwen2-Audio-ZS-CoT	66.7	56.3	54.4	49.7	42.8	54.0
Qwen2-Audio-CCoT-Emo	72.7(+7.2)	61.3(+6.2)	71.6(+14.5)	62.7(+9.8)	53.1(+6.7)	64.9(+9.1)
Qwen2.5-Omni [2]	64.7	57.0	59.3	54.3	50.1	57.0
Qwen2.5-ZS-CoT	65.0	57.9	55.6	47.9	45.4	54.4
Qwen2.5-CCoT-Emo	71.2(+6.5)	63.4(+6.4)	74.2(+12.3)	61.6(+7.3)	54.9(+4.8)	65.3(+8.3)
Kimi-Audio [3]	71.5	59.1	69.5	53.7	48.9	60.5
Kimi-Audio-ZS-CoT	69.4	60.8	67.6	47.1	46.6	58.3
Kimi-Audio-CCoT-Emo	78.4(+6.9)	64.9(+5.5)	76.6(+7.1)	63.4(+8.5)	56.4(+7.5)	67.7(+7.2)

Key Results: Consistent gains across all LALMs, improving zero-shot SER by up to 9.1%.

Ablation

Model	IEMOCAP	MELD	ESD	MERBench test1	MERBench test2	Overall
Qwen2.5-Omni-7B-CCoT-Emo	71.2	63.4	74.2	61.6	54.9	65.3
Qwen2.5-Omni-7B	64.7	57.0	59.3	54.3	50.1	57.0
w/out Acoustic Attribute	69.0	61.1	69.2	59.0	53.3	62.4
w/out Text Attribute	68.1	60.7	66.8	58.3	52.7	61.4
w/out Cross-Modal Relationship	70.0	61.9	71.8	60.2	53.9	63.6
w/ spectrograms featrue	67.6	59.8	63.5	56.1	50.4	59.4
w/ LALMs generate acoustic attribute	70.2	62.3	71.5	58.7	52.3	63.1
w/out JSON Format	70.1	62.2	73.0	59.1	54.8	63.9
Qwen-2.5-Omni-7B with free-form CoT	70.9	62.3	71.5	59.4	52.9	63.4
Qwen-2.5-Omni-3B	55.6	45.3	47.8	45.9	37.4	46.6
Qwen-2.5-Omni-3B-CCoT-Emo	59.8	51.1	53.0	50.2	40.8	50.9
128 Token Length	71.5	63.6	73.8	61.0	54.5	64.9
512 Token Length	70.7	62.6	74.0	61.2	54.3	64.6
1024 Token Length	70.4	62.5	74.1	61.2	54.2	64.4

Key Findings: Structured multimodal components jointly drive performance, with textual and acoustic cues most critical for accurate emotion reasoning